Diego Trevino

5/1/26

Assignment 3

CS420

Task 1 - Entity and Attribute Definition

For each of the six provided entities:

1. Clearly list all attributes.
2. Underline the primary key.
3. Identify composite attributes (if any).
4. Identify multivalued attributes.
5. State whether the entity is strong or weak and justify your answer.

1. Hospital

Type: Strong Entity. It is a strong entity because a hospital has a globally unique identifier (HospitalID) that doesn't depend on any other entity for identification. It exists independently.

Primary Key: <u>HospitalID</u>

Attributes: <u>HospitalID</u>, Name, Address, Phone.

Composite Attributes: None

Composite key: None

Multivalued attributes: None

2. Department

Type: Weak Entity. Department names are unique only within a hospital. The partial key is DeptName, and the full key is HospitalID + DeptName. A department cannot exist without its parent hospital, making it a weak entity.

Primary Key: <u>DeptName</u> + <u>HospitalID</u>

Partial Key: <u>DeptName</u>

Attributes: DeptName, Location, HospitalID
Composite Attributes: None
Multivalued attributes: None

3. Doctor

Type: Strong Entity. Each doctor has a unique DoctorID that does not rely on any other entity. A doctor works in exactly one department, but the doctor is still identified by DoctorID.
Primary Key: DoctorID
Attributes: DoctorID, Name, Specialty, DeptName, HospitalID
Composite Attributes: Name (FirstName, LastName)
Composite Key: None
Multivalued Attributes: None

4. Patient

Type: Strong Entity. Each patient has a unique PatientID that identifies them in the system. A patient can have appointments, but the patient does not depend on appointments for identification.
Primary Key: PatientID
Attributes: PatientID, Name, DOB, Phone
Composite Attributes: Name (FirstName, LastName)
Composite Key: None
Multivalued Attributes: None

5. Appointment

Type: Strong Entity. Appointment is modeled as a strong entity because it has its own unique AppointmentID. It involves exactly one patient and one doctor, but it does not need their keys to be identified.
Primary Key: AppointmentID
Attributes: AppointmentID, Date, Time, PatientID, DoctorID
Composite Attributes: None
Composite key: None

Multivalued attributes: None

6. Medication

Type: Strong Entity. Medication is a strong entity because each medication has a unique MedicationID. It can be prescribed in many appointments, but it exists independently as its own record.
Primary Key: MedicationID
Attributes: MedicationID, Name, Type
Composite Attributes: None
Composite key: None
Multivalued attributes: None

Task 2 - Relationship Identification

1. Hospital–Department

Relationship Name: Contains
Participating Entities: Hospital, Department
Cardinality: 1:N
Participation: Hospital = Partial, Department = Total
Relationship Attributes: None
Justification: One hospital can contain many departments, but each department belongs to exactly one hospital.

2. Department–Doctor

Relationship Name: Employs
Participating Entities: Department, Doctor
Cardinality: 1:N
Participation: Department = Partial, Doctor = Total
Relationship Attributes: None
Justification: A department may employ multiple doctors, but each doctor works in exactly one department.

3. Department–Head Doctor

Relationship Name: Heads
Participating Entities: Department, Doctor
Cardinality: 1:1
Participation: Department = Total, Doctor = Partial
Relationship Attributes: None
Justification: Each department has exactly one head doctor, but not every doctor is a department head.

4. Doctor–Doctor Supervision

Relationship Name: Supervises
Participating Entities: Doctor, Doctor
Cardinality: 1:N
Participation: Supervisor Doctor = Partial, Supervisee Doctor = Partial
Relationship Attributes: None
Justification: A doctor may supervise other doctors, but not every doctor supervises someone or has a supervisor.

5. Patient–Appointment

Relationship Name: Has
Participating Entities: Patient, Appointment
Cardinality: 1:N
Participation: Patient = Partial, Appointment = Total
Relationship Attributes: None
Justification: A patient may have many appointments, but each appointment involves exactly one patient.

6. Doctor–Appointment

Relationship Name: Handles
Participating Entities: Doctor, Appointment
Cardinality: 1:N
Participation: Doctor = Partial, Appointment = Total

Relationship Attributes: None
Justification: A doctor may have many appointments, but each appointment involves exactly one doctor.

7. Appointment–Medication

Relationship Name: Prescribes
Participating Entities: Appointment, Medication
Cardinality: M:N
Participation: Appointment = Partial, Medication = Partial
Relationship Attributes: Dosage, Duration
Justification: Dosage and duration belong to the relationship because they depend on a specific appointment and medication combination. The same medication can be prescribed in different appointments with different dosages or durations.

Task 3 — Weak Entity Analysis

Department

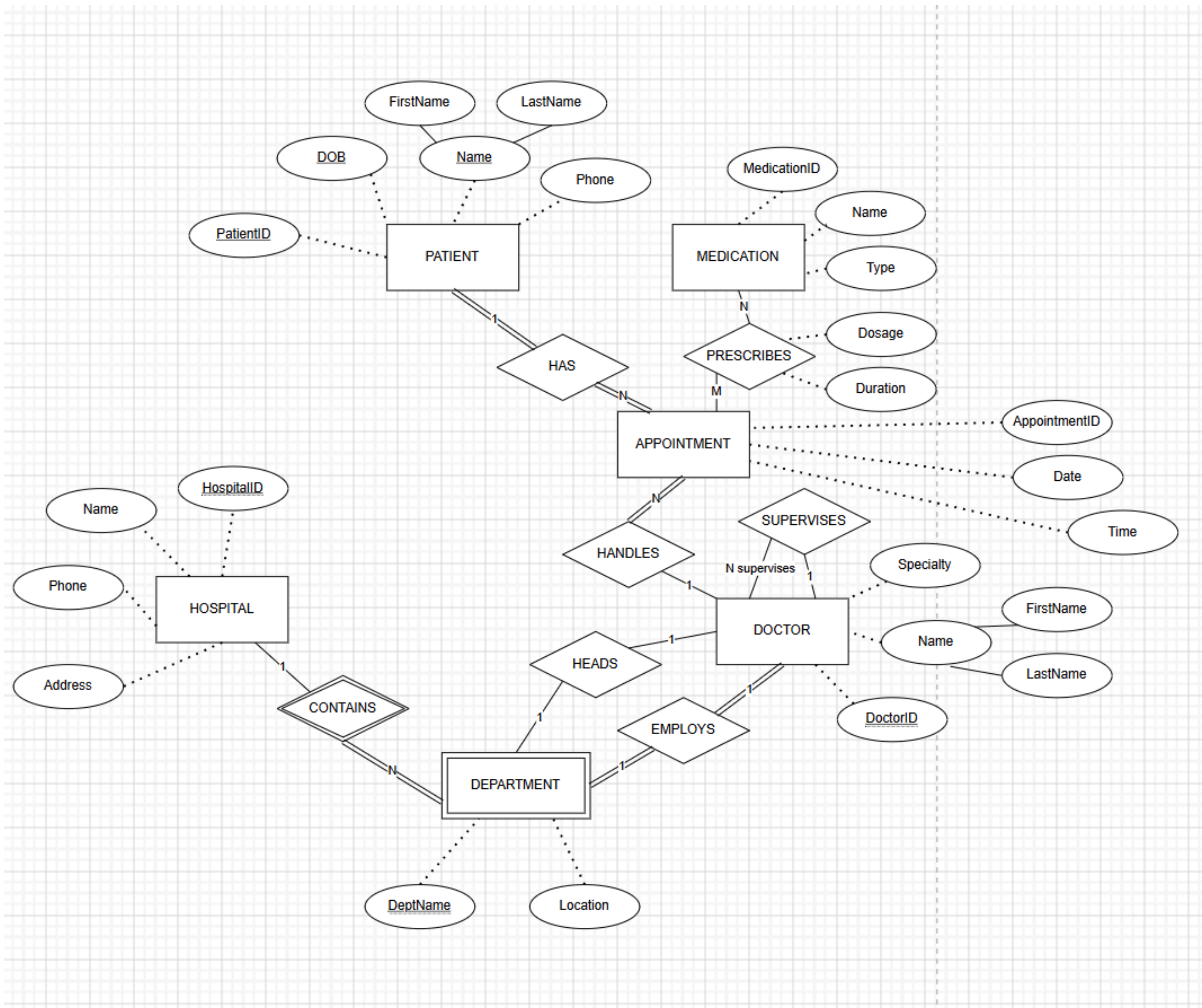
Department is a weak entity because its name alone isn't enough to identify it. Two different hospitals can both have a "Cardiology" department, so just knowing the department name doesn't tell you which one you're talking about. You need both the hospital ID and the department name together to actually identify it. On top of that, a department can't really exist without a hospital. If a hospital gets removed from the system, its departments don't mean anything on their own. Because of this, it also needs the identifying relationship with Hospital to form its full key. So since it doesn't have a unique key on its own and relies on Hospital to even exist, Department has to be modeled as a weak entity.

Appointment

Appointment is a strong entity because it has its own unique ID, AppointmentID, which is enough to identify it by itself. Unlike Department, it doesn't need to borrow anything from another entity to be identified. Even though every appointment is tied to a patient and a doctor, those connections are just foreign keys — they don't affect how the appointment itself gets identified. It also doesn't need an identifying relationship to exist in the system. So since it has its own primary key and doesn't depend on anything else for identification, Appointment should be modeled as a strong entity.

Task 4 - Final ER Diagram

Seen below.



Task 5 - Database Schema and Instance Examples

Hospital

HospitalID	**Name**	**Address**	**Phone**
H001	City General Hospital	Yakima, WA	509-555-0101
H002	Valley Medical Center	Ellensburg, WA	509-555-0202
H003	Pacific Coast Health	Seattle, WA	206-555-0303

Department

HospitalID	**DeptName**	**Location**
H001	Cardiology	Floor 3
H001	Emergency	Floor 1
H002	Cardiology	Floor 2

Doctor

DoctorID	**First Name**	**LastName**	**Specialty**	**HospitalID**	**DeptName**	**SupervisorID**
D001	Alice	Chen	Cardiologist	H001	Cardiology	NULL
D002	Bob	Martinez	Emergency Medicine	H001	Emergency	D001
D003	Carol	Smith	Cardiologist	H002	Cardiology	NULL

Patient

PatientID	**FirstName**	**LastName**	**DOB**	**Phone**
P001	Emma	Johnson	1990-03-15	509-555-1111
P002	Liam	Brown	1985-07-22	206-555-2222
P003	Sophia	Davis	2001-11-04	509-555-3333

Appointment

<u>AppointmentID</u>	Date	Time	PatientID	DoctorID
A001	2026-05-10	09:00	P001	D001
A002	2026-05-11	11:30	P002	D002
A003	2026-05-12	14:00	P003	D003

Medication

<u>MedicationID</u>	Name	Type
M001	Ibuprofen	Pain Reliever
M002	Amoxicillin	Antibiotic
M003	Lisinopril	Blood Pressure

DepartmentHead

Represents HEADS relationship.

<u>HospitalID</u>	<u>DeptName</u>	HeadDoctorID
H001	Cardiology	D001
H001	Emergency	D002
H002	Cardiology	D003

Prescription

Represents PRESCRIBES relationship.

<u>AppointmentID</u>	<u>MedicationID</u>	Dosage	Duration
A001	M001	400 mg	5 days
A001	M003	10 mg	30 days
A002	M002	500 mg	7 days

For the 1:N relationships, the foreign keys are included in the related tables. The DepartmentHead table represents the HEADS relationship, and the Prescription table represents the M:N PRESCRIBES relationship.